F-tests & Model Comparison

Revision sheet — Data Analysis for Psychology in R 2 (DAPR2), Week 4 · University of Edinburgh 2026–27

This week (= what can be examined)

- Test the **significance of the overall model** with an F-test.
- Use the same tool for **incremental tests**: does the model improve when I add variables?
- Compare **nested** models (incremental F-test) and **non-nested** models (AIC, BIC).

1. Is the model as a whole significant?

Question: does our **combination of x's** significantly improve prediction of y compared with having **no predictors**? Significant slopes or a high R^2 *suggest* it, but **do not directly test it**: that is the job of the **F-test**. In practice, evaluate a model by looking at **all** components together: significance of the βs, R^2 / adjusted R^2, and the F-test.

Test	Question	H_0
t-test of a β	Is **this** predictor useful (controlling for the others)?	$\beta_j = 0$
F-test (overall)	Is the model better than **nothing** (intercept-only)?	**all** $\beta_j = 0$
Incremental F-test	Does adding a **set** of predictors improve the model?	the added βs all = 0

2. The F-ratio

The F-ratio (F-statistic) compares **your model** with an **intercept-only model**, whose prediction for everyone is $\bar{y}$, i.e. predictors that explain no variance. **The more variance the predictors explain, the bigger F.**

$$\mathbf{F = MS_{Model} / MS_{Residual} = (SS_{Model} / df_{Model}) / (SS_{Residual} / df_{Residual})}$$

- **Mean squares (MS)** = sums of squares divided by their degrees of freedom: explained variation *per parameter* vs unexplained variation *per remaining df*.
- Reminder: $SS_{Total} = \Sigma(y_i - \bar{y})^2$, $SS_{Residual} = \Sigma(y_i - \hat{y}_i)^2$, $SS_{Model} = \Sigma(\hat{y}_i - \bar{y})^2$.
- If H_0 is true, $MS_{Model} \approx MS_{Residual}$ and **F ≈ 1**. More model than residual variation → **F > 1**.

3. Degrees of freedom

df = the number of values in a calculation that are **free to vary** if the result is held fixed. They depend on n and on the number of βs estimated.

Type	df	Why
Model	**k**	k + 1 βs are estimated (slopes + intercept); subtract 1 because not all can vary with the outcome held fixed.
Residual	**n − k − 1**	One df is lost for each of the k + 1 coefficients estimated from the data.
Total	**n − 1**	Based on y_i and $\bar{y}$: once $\bar{y}$ is fixed, all but one y can vary.

4. Testing F: the worked example

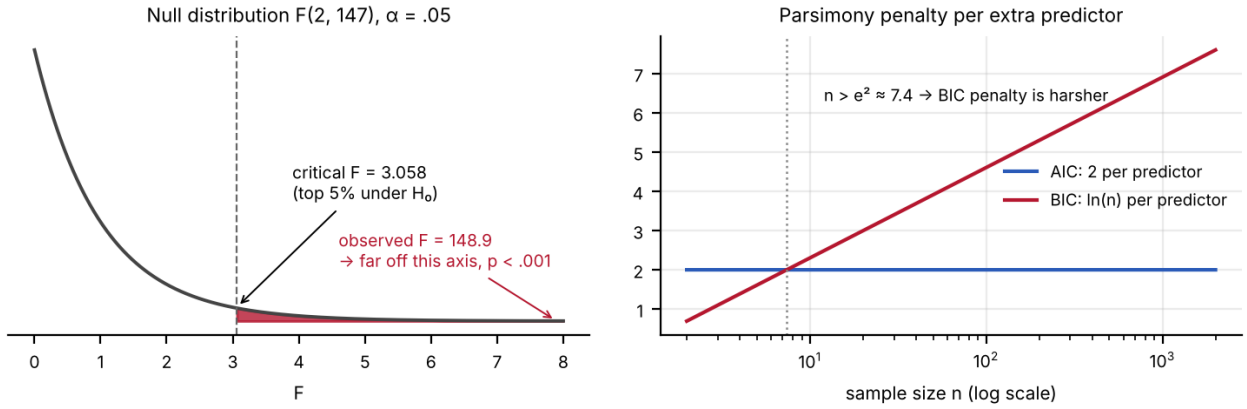
```
performance <- lm(score ~ hours + motivation, data = test_study2)
...
Residual standard error: 4.386 on 147 degrees of freedom
Multiple R-squared: 0.6696,     Adjusted R-squared: 0.6651
F-statistic: 148.9 on 2 and 147 DF, p-value: < 2.2e-16
```

Step	Example
1. Set α	α = .05
2. Read F	F = 148.9 (check by hand: (5729.23/2) / (2826.83/147) = 2864.6 / 19.23 ≈ 149)
3. Find the null distribution	F-distribution with **(df_{Model}, $df_{Residual}$)** = (k, n − k − 1) = (2, 150 − 2 − 1) = **(2, 147)**. Order matters: model df first.
4. Critical value	`qf(0.95, 2, 147)` = **3.058**
5. p-value	`1 - pf(148.9, 2, 147)` ≈ 0 → p < .001
6. Decide	148.9 > 3.058, p < α → **reject H_0** (all slopes = 0).

How to report it

Our model significantly predicted the variance in test score (F(2, 147) = 148.90, p < .001).

- The F-distribution changes shape with **both** df. F-tests are one-tailed (only large F counts against H_0).



Left: the rejection region of F(2, 147). Right: why BIC penalises extra predictors more than AIC for almost any sample size (see §7).

5. Model comparison

So far: is the model better than nothing (F-test)? Which variables are good predictors (t-tests)? Now: **when I change my model, does it improve?** E.g. is a model with x_1, x_2, x_3 better than one with just x_1?

Do NOT use model comparison to…	DO use it when…
• compare two models differing by **one parameter**: that parameter's t-test gives the same information. • **choose which predictors to include**: include those motivated by the RQ and needed to control confounds; don't drop predictors just because they are non-significant.	• you want to test the **combined impact of several predictors** (e.g. all five personality traits at once).

6. The incremental F-test (nested models)

The overall F-test is itself a comparison against an **empty (null) model** with only the intercept, which predicts ȳ for everyone (with no information, guessing the mean makes us "least wrong"). Empty model = all β = 0 = the F-test's H_0. The **incremental F-test** extends this to two models with fewer vs more predictors: it tests whether the **improvement in variance explained** from the added predictor(s) is significant.

$$\mathbf{F_{(df_R - df_F),\, df_F} = [\,(SSR_R - SSR_F) / (df_R - df_F)\,] / [\,SSR_F / df_F\,]}$$

F = full model (with the extra predictors), **R** = restricted model (without). SSR = residual sum of squares; df = **residual** degrees of freedom of each model.

Worked example (MIDUS2)

Data: MIDUS2 (n = 1,761). Outcome: self-rated health. Covariates: age, sex. Predictors: Big Five (O, C, E, A, N).

RQ: does personality predict self-rated health over and above age and sex?

Step 1 — fit both models: `m1 <- lm(health ~ age + sex, data = midus2)`

`m2 <- lm(health ~ age + sex + O + C + E + A + N, data = midus2)`

Step 2 — compare them: `anova(m1, m2)` (uses the **anova()** function, but this is **not** an ANOVA; the name is a coincidence).

	m1: age + sex	m2: + personality
R^2 / adjusted R^2	0.0046 / 0.0035	0.148 / 0.145
Overall F	F(2, 1758) = 4.09, p = .017	F(7, 1753) = 43.65, p < .001
Residual df	1758	1753

```
Model 1: health ~ age + sex
Model 2: health ~ age + sex + O + C + E + A + N
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1   1758 4740.2
2   1753 4055.4  5    684.85 59.208 < 2.2e-16 ***
```

By hand: $SSR_R - SSR_F$ = 4740.2 − 4055.4 = 684.85 ("Sum of Sq"); $df_R - df_F$ = 1758 − 1753 = **5** (the 5 traits added, "Df"). F = (684.85 / 5) / (4055.4 / 1753) = 136.97 / 2.313 = **59.2**, on (5, 1753) df.

How to report it

Personality was found to explain a significant amount of variance in self-rated health over and above the effects of age and sex (F(5, 1753) = 59.21, p < .001).

Nested vs non-nested

Nested ✓ (F-test allowed)	Non-nested ✗ (no F-test)

`m0: outcome ~ x1 + x2`
`m1: outcome ~ x1 + x2 + x3`
The predictors of one model are a **subset** of the other's.
	`m0: outcome ~ x1 + x2 + x4`
	`m1: outcome ~ x1 + x2 + x3`
	Each model has a **unique** variable (x4 vs x3).

Trap: missing data (NA)

An F-test requires models that are nested **and** fitted on **identical data**. lm() drops any row with an NA in y or any x in the model, so if the extra variables in the full model have missing values, the two models end up on **different rows**. Check for NAs (e.g. filter complete cases first).

7. AIC and BIC (nested or non-nested models)

$AIC = n \cdot \ln(SS_{Residual} / n) + 2k$ **$BIC = n \cdot \ln(SS_{Residual} / n) + k \cdot \ln(n)$**

- n = sample size, k = number of explanatory variables, ln = natural log.
- **Smaller (more negative) = better.** Choose the model with the smaller AIC or BIC.
- Do **not** require nested models (but still the same data).
- Both include a **parsimony correction**: they penalise complexity to avoid **overfitting** (adding predictors just to improve fit).
- **BIC penalises harder** whenever ln(n) > 2, i.e. n ≥ 8: essentially always.
- A single model's AIC/BIC is **meaningless**: only the **difference Δ** between models matters.

ΔBIC (Raftery, 1995)	Interpretation
< 2	No evidence of difference between models
2 – 6	Positive evidence of difference
6 – 10	Strong evidence of difference
> 10	Very strong evidence of difference

No equivalent thresholds exist for ΔAIC.

Comparison	AIC	BIC	Conclusion
Nested m1: age + sex m2: + O, C, E, A, N	m1: 6749.2 m2: **6484.5** Δ = 264.8	m1: 6771.1 m2: **6533.7** Δ = 237.4	m2 better; ΔBIC > 10 → very strong evidence (agrees with the F-test).
Non-nested nn1: O, C, E, A, N nn2: age + sex + PIL	nn1: **6501.5** nn2: 6565.0 Δ = 63.4	nn1: **6539.8** nn2: 6592.3 Δ = 52.5	nn1 (personality only) better; very strong evidence. No F-test possible here.

In R: `AIC(m1, m2)` and `BIC(m1, m2)`. The "df" column R prints counts all estimated parameters (βs + residual variance), e.g. 4 for age + sex; R's values include a constant, so they differ from the hand formula but **Δs are what count**.

8. Block 1 analysis workflow

Phase	Steps
1. Before fitting	Identify outcome and predictor variables → decide: coefficient significance tests or model comparison? → decide: transform (e.g. standardise) continuous predictor(s)?
2. Fit model	Write the mathematical model expression → formally define H_0 and H_1 → fit the model(s).
3. After fitting	Interpret coefficient estimates and/or the model comparison, as appropriate for the RQ.

Must remember

- **$F = MS_{Model}/MS_{Residual}$**, df = (k, n − k − 1); H_0: all slopes = 0; F ≈ 1 under H_0. Report F(df1, df2) = …, p = …
- **Incremental F** (`anova(m1, m2)`) compares **nested** models on the **same data**: does a **set** of predictors add explained variance?
- **AIC / BIC**: nested or non-nested; smaller is better; only Δ matters; BIC penalises complexity more.
- Don't use model comparison for a single parameter or to pick predictors.